

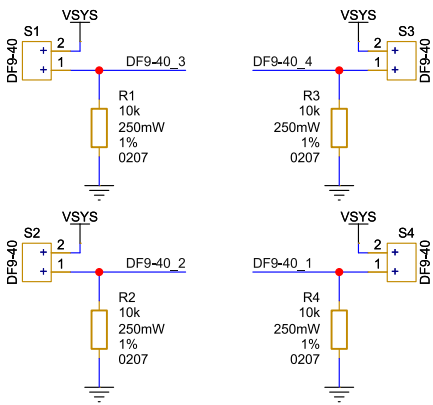
Vin max = 3V - 40V
Vout max = 1.5V - 35V
Output Imax = 2A continuous (3A peak)
Output ripple = 30mA

ALL GROUNDS MUST BE COMMON
NO 5V SIGNAL SHALL ENTER STM32 GPIO

LM339

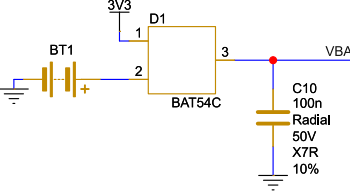
VCC = 2V - 36V
Input Vmax = VCC - 1.5V
Output type: Open collector (CANNOT source current)
Sink Imax = ~16mA (absolute max ~20mA)
MUST use pull-up resistor (4.7k-10k) on every output
DO NOT connect output directly to VCC
Input common-mode range: 0V to (VCC - 1.5V)
valid Vmax input ≈ 3.5V

Design Warning:
FSR divider MUST NOT exceed ~3.5V at input

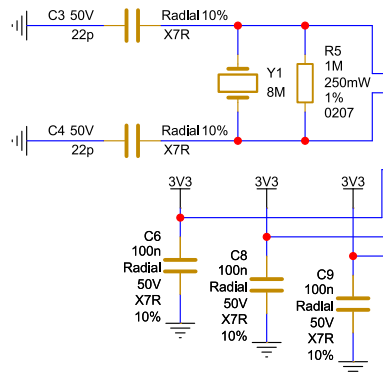


No load: >10MΩ
Pressed: ~1kΩ - 100kΩ
Recommended V = ≤5V

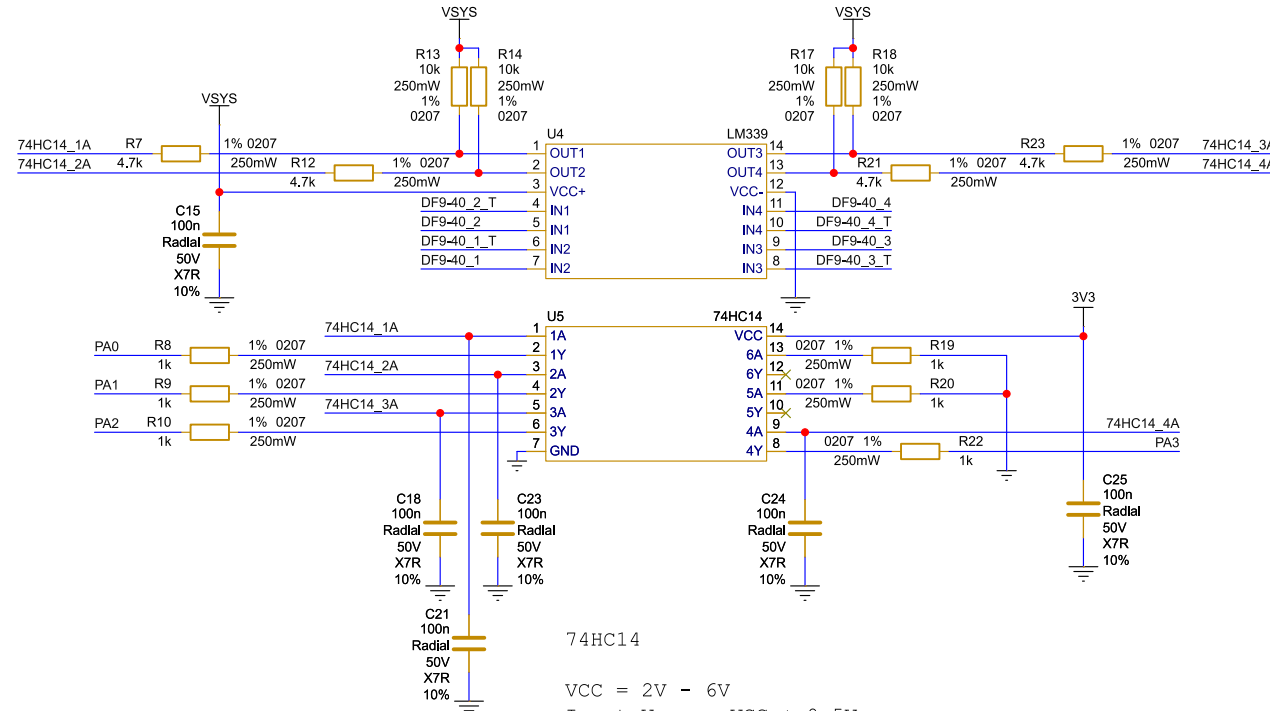
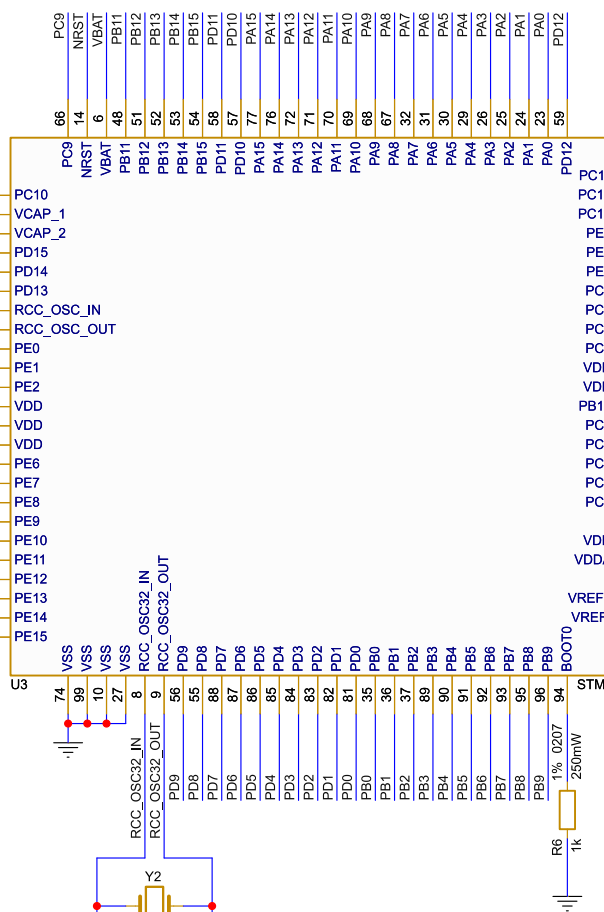
Design Warning:
High impedance = very noise sensitive
Avoid routing near motors or PWM lines



GPIO Vmax = 3.6V
GPIO Imax = ±8mA (recommended ≤5mA)
Use internal pull-down or pull-up in firmware



NOT 5V tolerant (most pins)
Direct 5V input = permanent damage

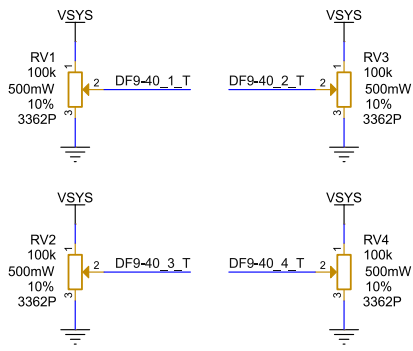


74HC14

VCC = 2V - 6V
Input Vmax = VCC + 0.5V
Output Imax = ±6mA typical (ABS max ~25mA total IC)

Schmitt thresholds:
VIH ≈ 0.6×VCC
VIL ≈ 0.3×VCC

Design Warning:
Never leave inputs floating
Input from LM339 must be limited. Use series resistor (4.7k)



Design Warning:
If wiper disconnects = floating input
Add (optional) 100k to GND for safety